

Improve Indoor Localization Accuracy by Enriching CSI Fingerprints with CDPM

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Abstract—Existing WiFi indoor localization methods require collecting a lot of Channel State Information (CSI) fingerprints at each reference point, which costs significant manpower and time. In this paper, we propose to expand CSI fingerprint dataset by using Conditional Diffusion Probabilistic Model (CDPM), generating artificial fingerprints for each reference positions additionally. The method avoids the high difficulty of adversarial training and mode collapse caused by traditional Generative Adversarial Network (GAN), and only needs to collect limited CSI fingerprints. The results on the public dataset show that the mean error of the proposed method is 2.7%~10.1% lower than the method based on ACGAN.

Keywords—WiFi indoor localization, fingerprint, CSI, CDPM

I. INTRODUCTION

With the development of localization service, the demand of indoor localization is also growing. Existing indoor localization methods are mainly based on Bluetooth^[1], Ultra-Wide Band^[2], ultrasound^[3], vision^[4] and so on. Among them, WiFi indoor localization has become a research hotspot because of its wide range and low cost. Compared with vision-based methods, WiFi indoor localization have no risk of privacy disclosure, so it can be used in hospitals, banks, nursing homes and other places with privacy requirement. However, WiFi indoor localization requires a large amount of fingerprints to build a fingerprint dataset, which costs a lot of manpower and time. Therefore, how to generate diverse fingerprints while improving indoor localization accuracy has important research value.

WiFi indoor localization generally uses Received Signal Strength (RSS) and CSI as fingerprints. The acquisition of RSS values is simple and the implementation complexity of the methods based on RSS is low. However, RSS has a fluctuation of about 5dB in a stable environment even, so the accuracy of indoor localization based on RSS is not high^[5-8]. On the contrary, with the technologies of Orthogonal Frequency Division Multiplexing and Multi-Input Multi-Output, CSI includes the channel information of all subcarriers on different antennas, which can better show

multi-path information. So the methods based on CSI is more accurate than those based on RSS. [9] proposed to use CSI as fingerprints for the first time. The authors used average amplitude values of CSI as fingerprints. But the method lost a lot of effective information, leading to low localization accuracy. In [10], an autoencoder composed of four hidden layers was used to reconstruct the amplitude values, and probabilistic method based on radial basis function was used to predict the estimated position. [11] used a neural network containing three convolutional layers and two fully connection layers to learn CSI fingerprints, and used SoftMax to output the estimated position. However, the above CSI localization methods need to collect a large number of fingerprints, and the neural network layer used is shallow, which can only deal with Line-of-Sight (LOS) scenes, but cannot deal with complex Non-Line-of-Sight (NLOS) scenes.

To improve the localization accuracy, the researchers proposed to enrich CSI fingerprints by using generative model. [12] proposed to use DCGAN to generate CSI amplitude data in view of the poor generation ability of WGAN, and replaced Adam optimizer with RMSProp optimizer to obtain more stable generation ability. [13] improved the discriminator of GAN by designing two neural networks with shared parameters to learn CSI data, giving GAN the ability of supervise generation. [14] input the reference position of CSI data into the discriminator in ACGAN, which was used to simultaneously learn the difference between the artificial data generated by the generator and the real data, as well as the categories of all data. However, the author used two regressors to predict estimated coordinates respectively, which wasted computing resources. Most of the above methods convert CSI data into the form of pictures for learning and generation, which could benefit from computer vision. Although the localization accuracy is improved, due to the limitation of GAN, there are defects such as high difficulty of adversarial training and mode collapse. In addition, the network structure needs to be adjusted according to different scenarios, which is not conducive to actual deployment.

Currently, using Diffusion Probabilistic Model (DPM) to enrich data is a hot research topic in the signal processing field. DPM can generate high-quality artificial data without adversarial training. DPM includes a diffusion process, a denoising process and a noise predictor. In the diffusion process shown in FIG. 1 from left to right, specific Gaussian

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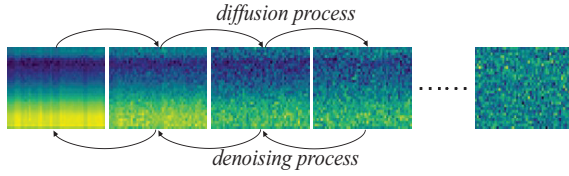


Fig. 1. Diffusion process and denoising process of DPM.

noise is constantly added to the original data to destroy the data. The noise predictor is used to learn the noise added in each step of the diffusion process. In the denoising process shown in Fig. 1 from right to left, DPM iteratively denoised a standard Gaussian noise to obtain artificial data similar to the original data by removing the noise predicted by noise predictor for each step in the diffusion process. [15] used DPM to generate unsupervised fake images, and the quality of generation is similar to Progressive GAN^[16], which verified the absolute generation ability of DPM in computer vision. In the research direction of image super resolution, [17] proposed to map the original data to the hidden space in lower dimensions and then use DPM for learning and generation, which achieved a balance between computing power and generation quality. [18] proposed a Conditional Diffusion Probabilistic Model (CDPM) without using classifiers, which made DPM to be a supervised generative model. In [19], speech signals containing noise are input into CDPM and non-noise speech signals are output, thus realizing speech enhancement.

In this paper, based on the above studies, we convert CSI amplitude data into multi-channel amplitude feature maps as fingerprints, and proposed to use CDPM instead of GAN to generate artificial CSI fingerprints. The method can adapt to different scenarios without adjusting the network structure, and avoid the problems of training difficulties and mode collapse of GAN. The reference positions are input into the conditional noise predictor, so that CDPM can generate the CSI fingerprints of the specified reference position. The method improves the localization accuracy when the original fingerprint dataset is limited. The effectiveness of the proposed method is verified by experiments of different indoor scenes. The main contributions of this paper are as follows:

- Slice CSI amplitudes using sliding window and convert the windows into multi-channel feature maps as fingerprints. Set specified sliding step to let fingerprints contain more channel information with limited dataset.
- Propose to expand CSI fingerprint dataset to improve indoor localization accuracy with CDPM, avoiding the difficulty of adversarial training and mode collapse brought by GAN.

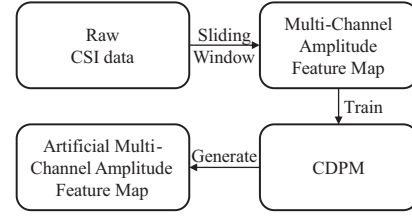


Fig. 2. The process of enriching CSI fingerprints by CDPM.

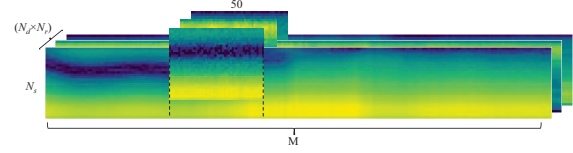


Fig. 3. Slice CSI data with sliding window.

- Experimental results show that compared with the existing schemes, the method using CDPM in this paper has higher localization accuracy

II. METHODS

This paper proposes a method to expand CSI fingerprint with CDPM. The method first converts the collected CSI data into multi-channel amplitude feature maps with sliding window as fingerprints, and then uses CDPM to generate a large number of artificial multi-channel amplitude feature maps for each reference position. The process of the method is shown in Fig. 2.

A. Multi-channel amplitude feature map

The CSI data collected actually are continuous time series. In order to ensure the real time of localization, it is necessary to divide the continuous CSI data into short time slices. In this paper, sliding window is used to slice CSI time series. The window size of the sliding window is set as 50 sampling periods like [10] and [20], and the sliding step is set as 25 sampling periods in order to extract more features from limited data.

Take a WiFi indoor localization system as an example. Assume the system has N_d transmit antennas and N_r receive antennas, forming $N_d \times N_r$ pairs of transmitting and receiving antennas, and N_s is the number of subcarriers. Collect M sampling periods of CSI for each reference position, then extract the amplitude data in collected CSI series, so each reference position has $N_d \times N_r \times M \times N_s$ amplitudes. After that, we use sliding window to segment the continuous CSI amplitude data to obtain $M/25$ windows, each window containing $N_d \times N_r \times 50 \times N_s$ amplitude measurements. Fig. 3 is the schematic diagram of sliding window slice.

Then, the windows are converted into multi-channel amplitude feature maps. Each feature map has $N_d \times N_r$ channels,

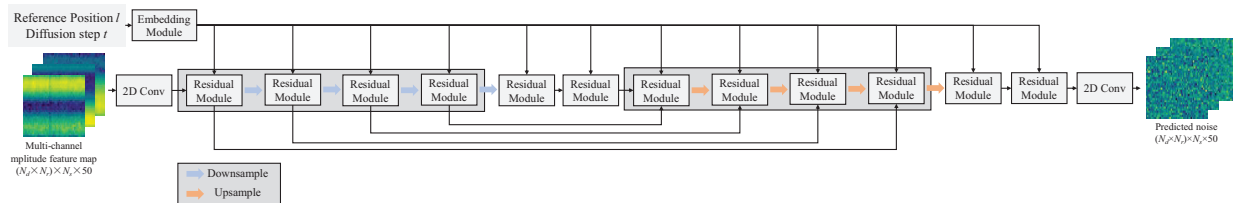


Fig. 4. Structure of Conditional Noise Predictor.

and the length and width of each channel is 50 and N_s . Denoting one feature map as $C = \{c_\varphi\}$, where $\varphi = 1, 2, \dots, N_d \times N_r$, c_φ is the amplitudes of 50 sampling periods corresponding of the N_s subcarriers on the φ_{th} pair of transmitting and receiving antennas. The feature maps of L different reference positions and their corresponding reference positions compose the original fingerprint dataset $P = \{(C^k, l^k)\}_{k=1}^K$, where C^k denotes the k_{th} multi-channel amplitude feature map in P , l^k is the corresponding reference position of C^k , $K = M/25 \times L$ is the total number of multi-channel amplitude feature maps in P .

B. Structure of CDPM

The CDPM proposed in this paper includes three parts: diffusion process, conditional noise predictor, and denoising process. The diffusion process continuously adds specific Gaussian noise to the original feature map in T steps of diffusion. The conditional noise predictor learns the noise added at each step in the diffusion process. The denoising process uses the noise predicted by the conditional noise predictor to iteratively denoise a gaussian noise data in reversed T steps to obtain artificial feature maps similar to the original feature maps.

(1) Diffusion Process

Denote C_t as the noise-containing multi-channel amplitude feature map after t steps of diffusion, where $t \in \{1, 2, \dots, T\}$. The specific Gaussian noise added in each step of the forward diffusion process is controlled by the noise factor β_t and changes with t . The change trend of β_t is the cosine scheme in [21]. The multi-channel amplitude feature map with noise after t steps of diffusion can be obtained by the following equation:

$$C_t = \sqrt{\alpha_t} C + \sqrt{1 - \alpha_t} \varepsilon_t, \quad \varepsilon_t \sim N(\mathbf{0}, \mathbf{I}) \quad (1)$$

where $\alpha_t = 1 - \beta_t$, $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$, ε_t is a Gaussian noise matrix

which is composed of sampling values of standard Gaussian distribution, with the same dimension as C . Using equation (1), the feature map containing noise after t steps of diffusion can be calculated directly from the original feature map C .

(2) Conditional Noise Predictor

In this paper, we improved the noise predictor in [15], inputting the reference position corresponding to the multi-channel amplitude feature map into the conditional noise predictor after embedding transformation. The conditional noise predictor is trained to predict the positive noise added at each step in the diffusion process. The conditional noise predictor is an improved neural network based on U-Net^[22], which mainly includes an embedding part, a downsampling part and an upsampling part. The result of the embedding part is input into all subsequent parts of the conditional noise predictor. This enables the conditional noise predictor to predict the noise according to the reference position and the number of diffusion step. The specific structure is shown in Figure 4.

The embedding part consists of a 2D convolution layer and an embedding module. The 2D convolution layer performs feature extraction of the feature maps, and the embedding

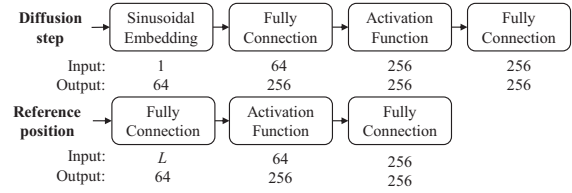


Fig. 5. Structure of embedding module.

module encodes the diffusion step number and the reference position. The embedding module encodes the number of diffusion steps and the reference position, and the network structure of it is shown in Fig. 5. The number of diffusion step t is input into Sinusoidal embedding layer^[23] and then input into two fully connected layers with 256 neurons. Meanwhile, the reference position is input two fully connected layers with 256 neurons. In addition, an activation function is added between two fully connection layers.

The downsampling part has four downsampling modules, and each downsampling module contains a residual module and a downsampling layer using maxpooling. The upsampling part has 4 upsampling modules. Each upsampling module contains a residual module and an upsampling layer based on nearest neighbor interpolation. When calculating the residual module before each upsampling layer, parallel the current result with the output of downsampling layer having the corresponding resolution. So, the upsampling part can use the information from the downsampling part. After the upsampling part, two residual modules and a 2D convolution layer are connected to recover the data to the original dimension.

The residual module contains two convolution modules, each convolution module consists of a 2D convolution layer, normalization layer and activation function, and the input of the residual module is added to the output of the second convolution module as output. In order to avoid the problem that the gradient of ReLU function is zero when the input is negative, we replace ReLU with SiLU, where

$$\text{SiLU}(x) = x \cdot \frac{1}{1 + e^{-x}} \quad (2)$$

The conditional noise predictor learns the noise added by each step of the input data in the diffusion process. In other words, the difference between the noise predicted by the conditional noise predictor and the noise added for each step of the diffusion process should be as small as possible. So the loss function of the conditional noise predictor is set as

$$\|\varepsilon_t - \tilde{\varepsilon}_\theta(C^k, t, l^k)\|^2, \quad \tilde{\varepsilon}_\theta = (1 + w)\varepsilon_\theta(C^k, t, l^k) - w\varepsilon_\theta(C^k, t) \quad (3)$$

where $\|\cdot\|^2$ is the L2 paradigm, ε_t is the Gaussian noise matrix added in the diffusion process. $\varepsilon_\theta(C_t^k, t, l^k)$ and $\varepsilon_\theta(C_t^k, t)$ is the noise predicted by the conditional noise predictor for the noise added in the diffusion process when the input with reference positions and without reference positions respectively. w is a weighting coefficient, which can control the diversity of data generated by CDPM. $\tilde{\varepsilon}_\theta$ is the weighted prediction noise, which is obtained by the weighted calculation of the above two predicted noises.

(3) Denoising Process

The denoising process also performs reversed T step denoising, using the noise predicted by the conditional noise

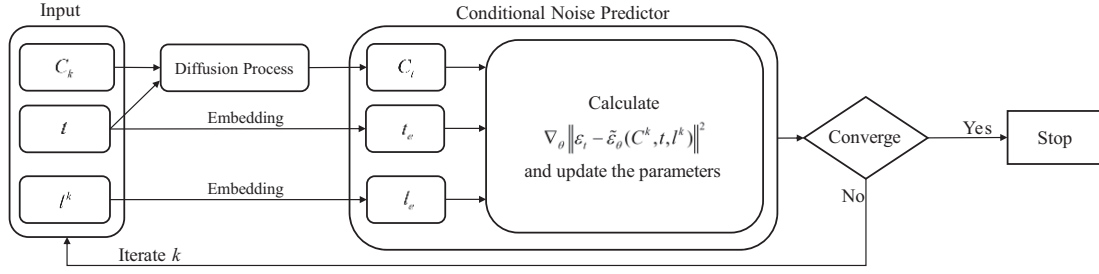


Fig. 6. Train conditional noise predictor with the diffusion process.

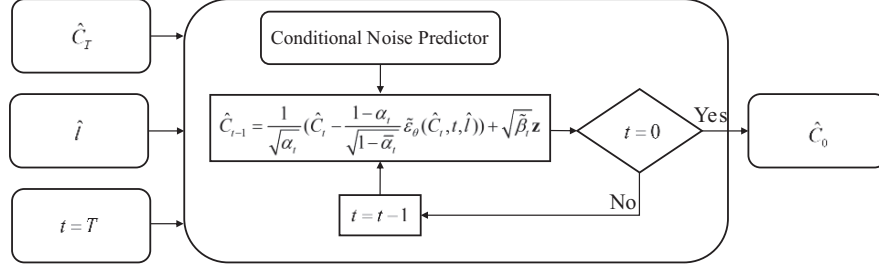


Fig. 7. Generate artificial feature map using the denoising process.

predictor to iteratively denoise a standard Gaussian noise matrix and recover it to an artificial data similar to the original data. The order of denoising processing is opposite to that of the diffusion process. In [15], the equation for recovering the noisy image from step t to step $t-1$ is as follows:

$$\hat{C}_{t-1} = \frac{1}{\sqrt{\alpha_t}} (\hat{C}_t - \frac{1-\alpha_t}{\sqrt{1-\alpha_t}} \varepsilon_{\theta}(\hat{C}_t, t)) + \sqrt{\tilde{\beta}_t} \mathbf{z} \quad (4)$$

where $\tilde{\beta}_t = \frac{1-\alpha_{t-1}}{1-\alpha_t} \beta_t$, $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I})$.

To obtain the artificial feature maps with reference positions, the noise predicted by conditional noise predictor $\tilde{\varepsilon}_{\theta}(\hat{C}_t, t, \hat{l})$ is used to replace the $\varepsilon_{\theta}(\hat{C}_t, t)$ in equation (4) as follows:

$$\hat{C}_{t-1} = \frac{1}{\sqrt{\alpha_t}} (\hat{C}_t - \frac{1-\alpha_t}{\sqrt{1-\alpha_t}} \tilde{\varepsilon}_{\theta}(\hat{C}_t, t, \hat{l})) + \sqrt{\tilde{\beta}_t} \mathbf{z} \quad (5)$$

C. Training and Generation of CDPM

(1) Training of CDPM

Assume that the diffusion process has T steps of diffusion. Use the diffusion process to train the conditional noise predictor, the process is shown in Fig. 6. The specific training steps are as follows:

1) Take a multi-channel amplitude feature map C^k and its reference position l^k in the original fingerprint dataset P . Take a random diffusion steps t , which is a random integer in $\{1, 2, \dots, T\}$;

2) Input the feature map containing accumulated noise obtained after t steps of diffusion together with l^k and t into the conditional noise predictor;

3) Use the embedding part in the conditional noise predictor to encode t and l^k , then replace l^k with an empty label with the probability λ ;

4) Use the result of equation (3) and the gradient descent method to update the neural network parameters of the conditional noise predictor:

5) Repeat steps 1) to 4) and iterate k , then update the neural network parameters of the conditional noise predictor until it converges.

(2) Generation of CDPM

Combined with the above reverse denoising process, the conditional noise predictor after training is used to generate artificial CSI multi-channel amplitude feature maps with a reference positions, the flow is shown in Fig. 7, and the specific steps are as follows:

1) Sample a standard Gaussian noise matrix $\hat{C}_T$ with the same dimension like C_k from standard Gaussian distribution;

2) Input $\hat{C}_T$ and $\hat{l}$ into the denoising process, and use equation (5) to obtain the artificial feature map $\hat{C}_0$;

3) Repeat steps 1) to 2) to obtain a large number of artificial feature maps whose reference positions are $\hat{l}$, and denote the set of these feature maps as Q_i ;

Using the above generation process to generate a set of artificial feature maps for different reference positions, and finally merged them into a large data set $Q = \{Q_i\}$, $\hat{l} \in L$, where L is a set composed of all reference positions. Merge P and Q as the expanded CSI amplitude feature map dataset.

III. EXPERIMENTS AND ANALYSIS

A. Experimental environment

In order to verify the effectiveness of the above method, this paper conducts comparative experiments on the public dataset [20]. The dataset contains two typical LOS and NLOS scenarios of a meeting room and a laboratory, which can well show the localization accuracy of different methods in different scenarios. The meeting room scene is a LOS scene

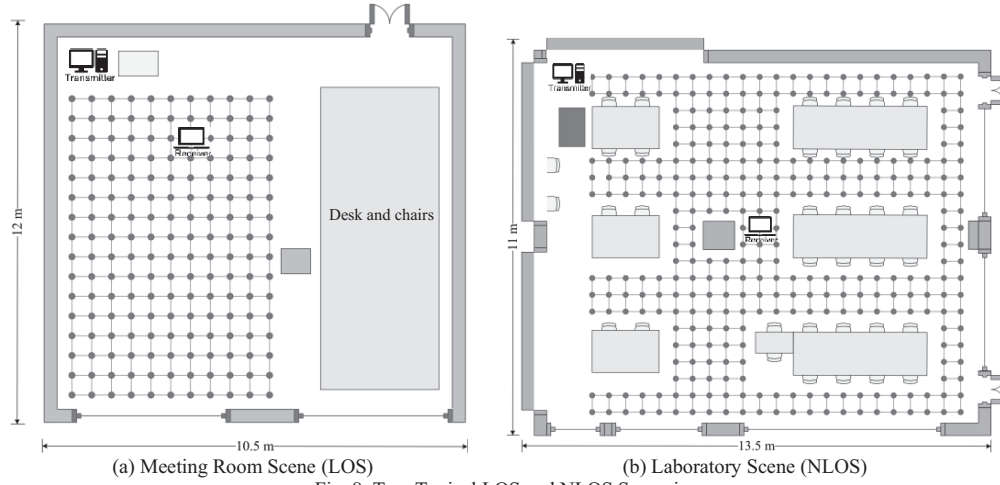


Fig. 8. Two Typical LOS and NLOS Scenarios.

TABLE I. POSITIONING ACCURACY COMPARISON IN THE MEETING ROOM SCENE

Generator	Regressor	100% Training Set		50% Training Set		20% Training Set	
		Mean Errors(m)	Std(m)	Mean Errors(m)	Std(m)	Mean Errors(m)	Std(m)
/	DeepFi[10]	2.673	1.692	3.212	1.921	3.821	2.021
/	ConFi[11]	2.769	1.776	3.155	1.858	3.743	1.966
/	Regressor[14]	0.828	1.358	0.934	1.397	1.509	1.985
ACGAN[14]	Regressor[14]	0.687	1.216	0.881	1.344	1.451	1.834
CDPM	Regressor[14]	0.618	1.113	0.829	1.320	1.412	1.703

TABLE II. POSITIONING ACCURACY COMPARISON IN THE LABORATORY SCENE

Generator	Regressor	100% Training Set		50% Training Set		20% Training Set	
		Mean Errors(m)	Std(m)	Mean Errors(m)	Std(m)	Mean Errors(m)	Std(m)
/	DeepFi[10]	4.067	2.991	4.921	3.629	6.131	3.224
/	ConFi[11]	4.207	3.192	4.818	3.494	6.274	3.475
/	Regressor[14]	1.747	2.359	1.870	2.382	2.240	2.517
ACGAN[14]	Regressor[14]	1.426	1.863	1.713	2.194	2.182	2.521
CDPM	Regressor[14]	1.312	1.799	1.627	2.072	2.114	2.423

as is shown in figure (a) in Fig. 8, there is no obstacles between the transmitting device and the receiving device. The size of the meeting room is 12m×10.5m, with 176 reference positions in total. The laboratory scene is an NLOS scene as is shown in figure (b) in Fig. 8, with a lot of obstacles like desk and chairs between the transmitting device and the receiving device. The size of the laboratory is 11m×13.5m, with 317 reference positions. One transmitting device and one receiving device are deployed for each scene. The transmitting device has 1 antenna and the receiving device has 3 antennas, each antenna pair has 30 subcarriers, and each reference position contains 1500 sampling periods of CSI amplitude data.

To simulate the localization accuracy in the actual scene, this paper divides each dataset into a training set and a test set with a ratio of 2:1. In other words, the first 1000 sampling periods are used for training, and the last 500 sampling periods are used for testing. By using sliding window proposed in this paper, the CSI amplitude data in training set and the test set are divided into datasets composed of multi-channel amplitude feature maps and reference positions. In addition, to better simulate the real scene, the starting index of the sliding window in the test set are non-repeating random value.

This paper uses Pytorch1.13 as the deep learning framework, and uses Python 3.9 to write all programs, running on a RTX3090.

B. Localization accuracy analysis

In order to verify the effectiveness of the method in this paper in different scenes, 100%, 50%, and 20% of the data in the training set were used respectively to compare the method proposed in this paper with previous methods^[10,11,14]. The total number of diffusion and denoising steps of CDPM is set to 1000, the learning rate is set to 0.0001, train epochs is set to 5000, λ is set to 0.2, w is set to 6. All generation methods generated 16 artificial fingerprints for each reference position. When calculating the localization accuracy, use the regressor in [14] to train the expanded dataset for 200 epochs after the original data set and the simulated data set are merged, and set the learning rate of the regressor to 0.001. All models are trained with a batch size of 128 and use Adam optimizer.

In the meeting room scene, the mean errors and standard deviations of different methods are shown in Table 1. The mean errors of the method proposed in this paper on the 100%, 50%, and 20% training sets are 0.618m, 0.829m, and 1.412m, which are 10.1%, 5.9% and 2.7% higher than ACGAN. In the laboratory scene, the mean error and standard deviation of different methods are shown in Table 2. The mean errors of the method proposed in this paper on the 100%, 50%, and 20% training sets are 1.312m, 1.627m and 2.114m, which are 8.0%, 5.1%, and 3.2% higher than ACGAN.

TABLE III. TIME CONSUMPTION COMPARISON IN THE METTING ROOM SCENE WITH 50% TRAINING SET

Methods	Generation	Regressor Training	Total Time
Regressor[14]	/	0.35h	0.35h
ACGAN[14]	6.47h	0.63h	7.10h
CDPM	8.21h	0.63h	8.84h

TABLE IV. TIME CONSUMPTION COMPARISON IN THE LABORATORY SCENE WITH 50% TRAINING SET

Methods	Generation	Regressor Training	Total Time
Regressor[14]	/	0.53h	0.53h
ACGAN[14]	9.92h	0.96h	10.88h
CDPM	13.31h	0.96h	14.27h

According to the above results, the localization accuracy of the method proposed in this paper is higher than that of ACGAN on different degrees of missing training sets, which proves the effectiveness of the method proposed in this paper. In addition, compared with the positioning method that does not use the generative model, the localization accuracy of the method in this paper can be greatly improved in both LOS and NLOS scenarios, and only 50% of the training set data is needed to achieve similar localization accuracy as using 100% training set.

The time consumption of different localization methods with 50% training set is shown in Table 3 and Table 4. Compared with traditional methods, generative models require additional time to generate artificial data. The increase of data in training sets further leads to the increase of the time consumption while training regressor. The localization method based on CDPM can provide higher localization accuracy at the cost of longer training time. Specifically, CDPM costs 26.89%~34.17% more time in the generation phase than ACGAN. But in the online stage, the time consumption is almost the same because the regressor used and the size of input are the same.

The accuracy improvement brought by CDPM has a marginal effect. The trend of the increase in the localization mean error in different scenarios is shown in Fig. 9. Compared with the method without using the generative model, it can be seen that the less data in the training set, the less the localization accuracy is improved. This is because the redundant information provided from a small amount of original data is limited. It is difficult for the generative model to generate artificial data that conforms to the distribution of actual deployment environment with limited original data. The artificial data that not conforms to the distribution of actual deployment environment may also cause the increase of standard deviation of the localization accuracy.

IV. CONCLUSION

This paper proposes a method of combining CSI with CDPM to improve indoor localization accuracy. Using CDPM saves the manpower and time required for collecting CSI data. First, the collected CSI data is converted into a multi-channel amplitude feature map, and then expanded with CDPM. Compared with the traditional GAN-based generation model, CDPM avoids the problems of training difficulties and mode collapse, and can generate more diverse CSI data. By using CDPM, the number of samples in the training set is effectively increased. In this paper, experiments are carried out on different degrees of missing training sets to verify the proposed method. The results show that CDPM can enhance

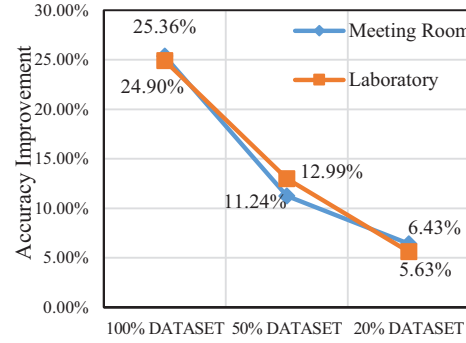


Fig. 9. Accuracy improvement on different degrees of missing datasets.

the diversity of original data, thereby improving the localization accuracy in the case of limited training set data.

REFERENCES

- [1] Chen J, Zhou B, Bao S, et al. A data-driven inertial navigation/Bluetooth fusion algorithm for indoor localization[J]. IEEE Sensors Journal, 2021, 22(6): 5288-5301.
- [2] Rey R, Corzetto M, Cobano J A, et al. Human-robot co-working system for warehouse automation[C]//2019 24th IEEE International Conference on Emerging Technologies and Factory Automation (ETFA). IEEE, 2019: 578-585.
- [3] Gabbrielli A, Xiong W, Schott D J, et al. An echo suppression delay estimator for angle-of-arrival ultrasonic indoor localization[J]. IEEE Transactions on Instrumentation and Measurement, 2021, 70: 1-12.
- [4] Morar A, Moldoveanu A, Mocanu I, et al. A comprehensive survey of indoor localization methods based on computer vision[J]. Sensors, 2020, 20(9): 2641.
- [5] Yang Z, Zhou Z, Liu Y. From RSSI to CSI: Indoor localization via channel response[J]. ACM Computing Surveys (CSUR), 2013, 46(2): 1-32.
- [6] Youssef M, Agrawala A. The Horus WLAN location determination system[C]//Proceedings of the 3rd international conference on Mobile systems, applications, and services. 2005: 205-218.
- [7] Sangthong J, Thongkam J, Promwong S. Indoor wireless sensor network localization using RSSI based weighting algorithm method[C]//2020 6th International Conference on Engineering, Applied Sciences and Technology (ICEAST). IEEE, 2020: 1-4.
- [8] Thaljaoui A, Val T, Nasri N, et al. BLE localization using RSSI measurements and iRingLA[C]//2015 IEEE international conference on industrial technology (ICIT). IEEE, 2015: 2178-2183.
- [9] Xiao J, Wu K, Yi Y, et al. FIFS: Fine-grained indoor fingerprinting system[C]//2012 21st international conference on computer communications and networks (ICCCN). IEEE, 2012: 1-7.
- [10] Wang X, Gao L, Mao S, et al. DeepFi: Deep learning for indoor fingerprinting using channel state information[C]//2015 IEEE wireless communications and networking conference (WCNC). IEEE, 2015: 1666-1671.
- [11] Chen H, Zhang Y, Li W, et al. ConFi: Convolutional neural networks based indoor Wi-Fi localization using channel state information[J]. IEEE Access, 2017, 5: 18066-18074.
- [12] Li Q, Qu H, Liu Z, et al. AF-DCGAN: Amplitude feature deep convolutional GAN for fingerprint construction in indoor localization systems[J]. IEEE Transactions on Emerging Topics in Computational Intelligence, 2019, 5(3): 468-480.
- [13] Chen K M, Chang R Y. Semi-supervised learning with GANs for device-free fingerprinting indoor localization[C]//GLOBECOM 2020-2020 IEEE Global Communications Conference. IEEE, 2020: 1-6.
- [14] Wei W, Yan J, Wan L, et al. Enriching indoor localization fingerprint using a single AC-GAN[C]//2021 IEEE Wireless Communications and Networking Conference (WCNC). IEEE, 2021: 1-6.
- [15] Ho J, Jain A, Abbeel P. Denoising diffusion probabilistic models[J]. Advances in Neural Information Processing Systems, 2020, 33: 6840-6851.

- [16] Karras T, Aila T, Laine S, et al. Progressive growing of gans for improved quality, stability, and variation[J]. arXiv preprint arXiv:1710.10196, 2017.
- [17] Rombach R, Blattmann A, Lorenz D, et al. High-resolution image synthesis with latent diffusion models[C]//Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2022: 10684-10695.
- [18] Ho J, Salimans T. Classifier-free diffusion guidance[J]. arXiv preprint arXiv:2207.12598, 2022.
- [19] Lu Y J, Wang Z Q, Watanabe S, et al. Conditional diffusion probabilistic model for speech enhancement[C]//ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). IEEE, 2022: 7402-7406.
- [20] Zhu X, Qiu T, Qu W, et al. BLS-location: A wireless fingerprint localization algorithm based on broad learning[J]. IEEE Transactions on Mobile Computing, 2021.
- [21] Nichol A Q, Dhariwal P. Improved denoising diffusion probabilistic models[C]//International Conference on Machine Learning. PMLR, 2021: 8162-8171.
- [22] Li X, Chen H, Qi X, et al. H-DenseUNet: hybrid densely connected UNet for liver and tumor segmentation from CT volumes[J]. IEEE transactions on medical imaging, 2018, 37(12): 2663-2674.
- [23] Vaswani A, Shazeer N, Parmar N, et al. Attention is all you need[J]. Advances in neural information processing systems, 2017, 30.